

DAY 3 — Sorting & Aggregates

Analyze data like a business owner and learn how companies actually use databases.

```
SELECT COUNT(*), AVG(price)
FROM mobile_phones
GROUP BY brand;
```


The Database Journey So Far



Day 1

Store Data

Learning how to create tables and insert raw information securely into the database.



Day 2

Find Data

Using queries to search, filter, and extract specific records based on simple conditions.



Day 3 (Today)

Analyze Data

Grouping, calculating averages, finding minimums/maximums, and sorting final results.

Real-Life Example: Mobile Shop

Imagine you own a modern mobile phone shop. Looking at rows of raw data won't help you make decisions. You want answers to business questions:

- > Which phone is the **most expensive**?
- > What is the **average price** of your inventory?
- > **How many** phones do you have per brand?
- > Which brand **sells the most**?

SQL helps you calculate these automatically, transforming raw data into business intelligence.



1. Sorting Results

Arranging data systematically from lowest to highest, or highest to lowest, to easily identify top performers or most expensive items.

Using ORDER BY

The ORDER BY clause is used to sort the result-set in ascending or descending order based on one or more columns.

Ascending Order (Default)

Sorts data from Small → Big, A → Z, or Oldest → Newest.

```
SELECT * FROM students  
ORDER BY marks;
```

Use case: Finding the cheapest products or oldest records.

Descending Order (DESC)

Sorts data from Big → Small, Z → A, or Newest → Oldest.

```
SELECT * FROM students  
ORDER BY marks DESC;
```

Use case: Finding top students, highest salaries, or most expensive items.

2. Aggregate Functions (The Calculator)

These functions work on many rows to return a single calculated result.

FUNCTION	MEANING	REAL LIFE EXAMPLE	SQL EXAMPLE
COUNT()	Total number of rows	Total students in school	SELECT COUNT(*) FROM students;
SUM()	Total combined value	Total sales revenue	SELECT SUM(price) FROM sales;
AVG()	Average value	Average class marks	SELECT AVG(marks) FROM students;
MAX()	Highest value	Highest phone price	SELECT MAX(price) FROM products;
MIN()	Lowest value	Lowest student marks	SELECT MIN(marks) FROM students;

3. GROUP BY Clause (Very Important ★)

The Concept

Make groups out of your data and calculate an aggregate function inside each specific group.

Real-Life Example: A teacher doesn't just want the overall school average. They want the average marks *per class*.

Expected Output:

class	avg_marks
5th	85
6th	75

The Syntax

```
-- General Syntax
SELECT column, FUNCTION(column)
FROM table
GROUP BY column;
```

The Example

```
-- Average marks per class
SELECT class, AVG(marks)
FROM students
GROUP BY class;
```


4. HAVING Clause & Memory Trick

The Problem: The WHERE clause cannot filter grouped results (aggregates). So we use the HAVING clause.

Example: Show only classes where average marks > 80.

SELECT
GROUP BY

Step 2

Make groups

GROUP BY

students
> 80;

Step 1

Filter individual rows

WHERE

Step 3

Filter groups

HAVING

Hour 2: Practice

Mobile Shop Challenge 📱

It's time to put your analyst skills to the test using the mobiles table.

Write SQL queries to find:

- Average price per brand
- Most expensive phone in stock
- Total number of phones in stock
- Brands with an average price > 20000



Students Table Practice Tasks



Task 1: Sorting

- Sort marks (low → high)
- Sort marks (high → low)
- Sort alphabetically by name



Task 2: Aggregates

- Count total students
- Find average marks
- Find highest marks
- Find lowest marks



Task 3 & 4: Grouping

- Number of students per class
- Average marks per class
- Classes where avg > 80

Real Business Understanding

Companies use these exact SQL concepts daily. You are now learning real Data Analyst skills.

COMPANY / INDUSTRY	REAL-WORLD SQL USE CASE	KEY CONCEPT USED
 Amazon	Finding average product sales by category	GROUP BY category, AVG(sales)
 School Systems	Tracking overall class performance and top students	ORDER BY marks DESC, AVG(marks)
 Banks	Calculating total transaction volumes per branch	GROUP BY branch, SUM(amount)
 Hospitals	Counting patient statistics per ward	GROUP BY ward, COUNT(patient_id)

Day 3 Completed! 🎉

You can now sort data, calculate totals, analyze groups, and think like an analyst.

🔥 Tomorrow — DAY 4 (Big Topic)

You'll learn how to connect databases like relationships between people:

- ✓ INNER JOIN
- ✓ LEFT JOIN
- ✓ RIGHT JOIN
- ✓ FULL OUTER JOIN

When practice is done, say: "**Day 3 Completed**"

Image Sources



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